Fiscal Year:	FY 2004	**Task Last Updated:**	FY 02/23/2006
PI Name:	Benton, Eric Ph.D.		
Project Title:	Space Radiation Shielding Using the NASA Deep Space Test Bed		
Division Name:	Physical Sciences		
Program/Discipline:	RADIATION SHIELDING		
Program/Discipline--Element/Subdiscipline:	RADIATION SHIELDING--Radiation shielding		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	San Rafael	**State:**	CA
Zip Code:	94915-0788	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Radiation Biology and Shielding 03-OBPR-07
Start Date:	06/01/2004	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Allen, Frank (Eril Research Inc) Gersey, Brad B (Prairie View A&M) Benton, Eugene V (Eril Research Inc)		
Grant/Contract No.:	NNM04AA75G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We propose an extensive, cost effective research program to characterize the radiation shielding properties of baseline and novel multifunctional materials during exposure to the free-space galactic cosmic ray (GCR) environment aboard the NASA Deep Space Test Bed (DSTB). Measurements behind shielding will be made with instrumentation currently used for astronaut dosimetry aboard low-earth orbit (LEO) spacecraft. The proposed research program will result in 1) measured data sets from baseline shielding materials for use in validating and testing of space radiation transport codes currently being developed by NASA, 2) characterization of the shielding properties of novel multifunctional materials currently being developed, and 3) maximum scientific benefit from the existing DSTB radiation detection instrument suite. The DSTB instrument suite includes CR-39 plastic nuclear track detectors (PNTD), thermoluminescent dosimeters (TLD), two NASA tissue equivalent proportional counters (TEPC), and Liulin-4 portable Si spectrometers. Measurements will include LET/lineal energy spectrum, total absorbed dose, dose equivalent, and average quality factor behind shielding materials of varying composition and thickness. Shielding materials include baseline materials such as aluminum and polyethylene, and novel multifunctional materials including hydrogen-doped carbon composites, thermoplastics and metal hydrides. The proposed program is a logical extension of ground-based research projects currently underway using this same instrumentation and shielding at heavy ion accelerator facilities.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New project for FY2004.
Bibliography Type:	Description: (Last Updated: 11/13/2025)